MODE-S Wind on a phone

The mobile view `/app`: Skew-T, time–height section, ILS approach and approach conditions in your pocket

Otso Laakso (OH2GAX)
Tuusula, Finland · near Helsinki-Vantaa airport (EFHK)

In short. The full MODE-S Wind pages are made for a computer screen. The mobile view is a light page for phones: open the address of MODE-S Wind with `/app` at the end. It shows *one product at a time*, filling the screen:

- **Skew-T · Area now** — the temperature and wind profile measured by the aircraft over the last 30 minutes to 3 hours, with the radiosonde and the MEPS model if you like;
- **Skew-T · Area hour** — any stored hourly profile;
- **Time–height** — the hourly profiles side by side: fronts, inversions and jet streams as they pass;
- **ILS approach** — the aircraft on final on the 3° glideslope, live, with the wind measured along the approach and any windshear alert;
- **ACI · WS · ST** — the approach conditions index and windshear of every landing and departure, or the space–time diagram of the wind changes, with the flights of the period as a list;
- **History** — the landings or the departures of 1 hour to a day, or of any date, as cards with their winds per height band and their reports, with a search.

Everything is chosen from the ≡ menu at the top left. Tap a diagram for its values. *Add to Home Screen* gives MODE-S Wind its own icon, and it opens full screen like an app. The diagrams are the same as on the full site, so the other user guides explain how to read them; this guide explains how to use them on a phone.

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1 Getting started

1.1 Opening the mobile view

Open the phone's browser and go to the address you use for MODE-S Wind on a computer, with `/app` added at the end — for example `http://your-address/app`. The phone has to reach MODE-S Wind the same way a computer does (your own network, or the connection set up for it). The browser asks for the same user name and password as the other pages and normally remembers them.

The mobile view opens the product you looked at last. All choices — the view, its options, the theme — are remembered on the phone. The Skew-T options and the theme are the same settings as on the Live Map and the Sounding page *of the*

same browser: if you also open the full site on the phone, it starts with the same choices.

1.2 Add to Home Screen

Android (Chrome): open `/app`, tap the browser menu ⋮ and choose Add to Home screen (on some phones Install app), then Add.

iPhone and iPad (Safari): open `/app`, tap the Share button (the square with an arrow) and choose Add to Home Screen, then Add.

The MODE-S Wind icon (right) appears among your apps. It opens the mobile view full screen, without the browser's address bar, and can be switched to like any other app. The first time it is opened the phone may ask for the user name and password again.



the icon

1.3 The screen

A slim bar at the top holds three things:

- ≡ at the left opens the menu (Section 2);
- the **name of the view** and under it a short line: what is shown and when it was updated — e.g. `last 60 min · 15:55 UTC`, `24.10.2026 15–16 UTC`, or on the ILS view the number of aircraft on the approach, the QNH and any active windshear alert;
- a **dot** at the right: the system health of MODE-S Wind, the same indicators as on the Statistics page. *Green*: all parts work. *Amber*: something is late, retrying or using a fallback (e.g. no new MEPS model data). *Red*: a part has stopped. The menu's General section says which part; the Statistics page on the full site has the details.

Below the bar is the product. All times are UTC.

2 The menu

Tap ≡. The menu slides in from the left (Fig. 1); × or a tap beside it closes it.

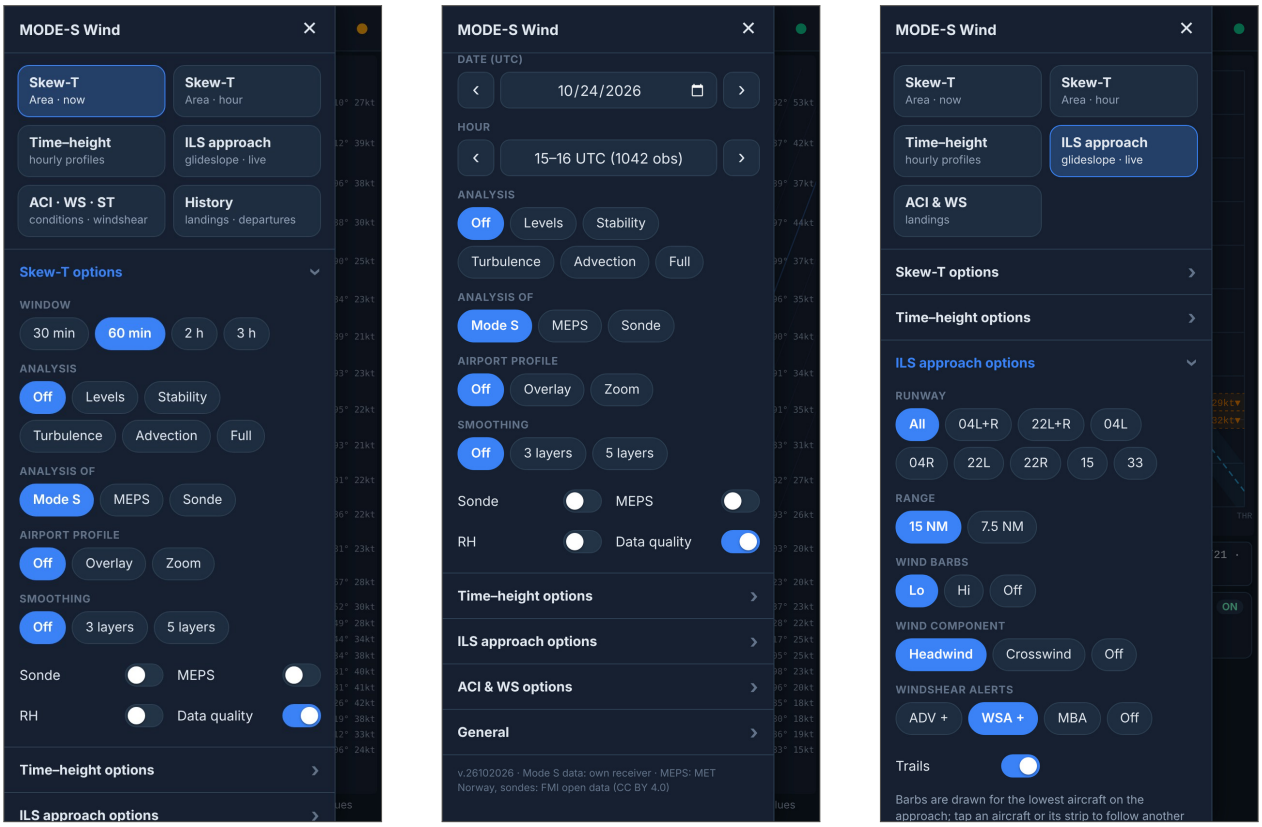


Figure 1. The menu. Left: the six views as tiles (the shown one highlighted) and the options of the shown view, here Skew-T, opened below them. Middle: Area · hour with its date and hour steppers and the switches. Right: the ILS approach options; the other products' options are folded but one tap away.

Views. The six tiles at the top choose the product. A tap switches to it and closes the menu.

Options. Below the tiles each product has a folding section: Skew-T options, Time–height options, ILS approach options, ACI · WS · ST options and History options. The shown view's section opens by itself (its title in blue); the others stay folded and open with a tap, so you can, for example, set the Skew-T options while the History is shown. The menu scrolls on its own.

Controls. Choices are buttons: the highlighted one is in use, a tap on another changes it at once. On / off settings are switches. Dates have a calendar field and, for stored hours, ⟨ ⟩ buttons that step one day or one hour back and forward (over midnight to the next or previous day). Changes take effect immediately; there is no *Apply* button.

General. The theme (Dark / Light), the system health in words, About and User guides (from 28 October 2026, Fig. 2), a link to the full site and to this guide. About shows the same text as the About window of the full site: who made MODE-S Wind and how to reach him, the receiver site, how the observations are made, the products, the windshear detection and the credits. User guides lists all the guides of the full site — e.g. the Windshear guide or the windshear study log — with their version, date, size and contents (Contents); Open PDF opens one in the phone's PDF viewer, Download saves it. Both open over the view; × or the phone's *Back* closes them.

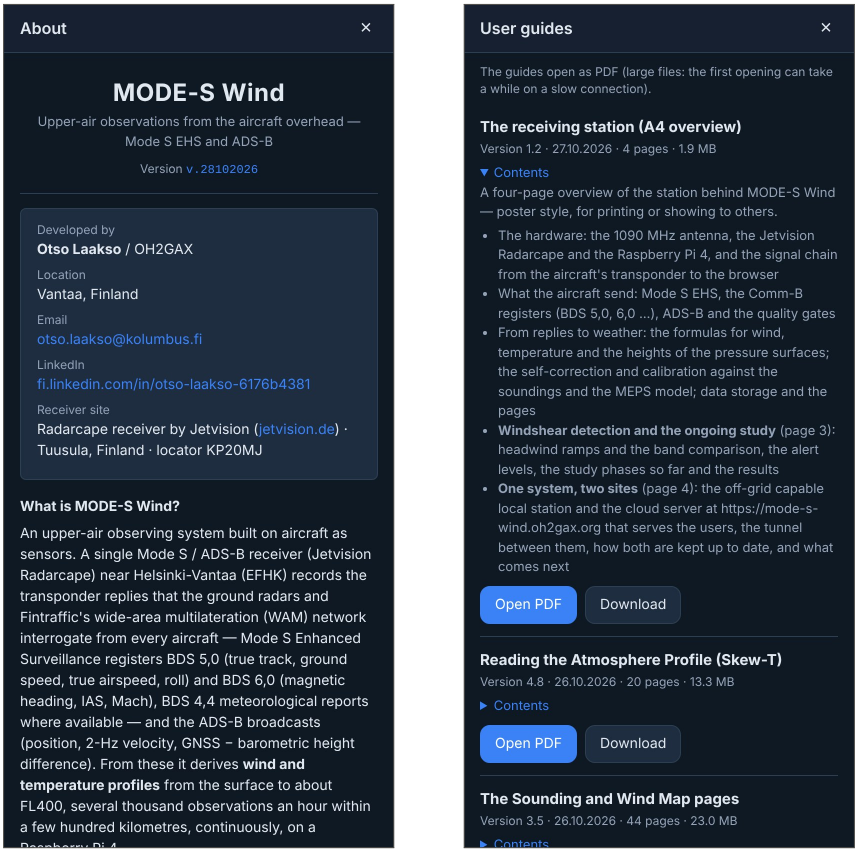


Figure 2. About (left) and User guides (right, with the contents of the first guide opened), from the menu's General section.

3 Touch, rotation and updates

Tap for values. On the full site the values appear when the mouse is over a diagram. On the phone, *tap* the diagram: a box shows the values at that point — on the Skew-T the level's temperature and wind (and the sonde's and model's), on the time–height section the hour and layer, on the ACI & WS chart the nearest landing, departure or go-around, on the space–time diagram the flights of that cell. *Drag* sideways or up and down to move along; tap elsewhere to move the box. On the ILS view a tap on an aircraft chooses whose wind barbs are drawn (Section 6).

Scrolling. When a view is taller than the screen — ACI · WS · ST with its list of flights, the History, the ILS view with many aircraft, or a Skew-T turned to landscape — swipe up and down to scroll, also when the swipe starts on the diagram. A short tap still gives the values.

Portrait and landscape. In portrait each diagram takes the full width and as much of the height as it needs. Turn the phone for more room across: the time–height section gets a longer time axis, the ILS profile spreads over the screen (Fig. 3); the Skew-T keeps its shape and the page scrolls.



Figure 3. Landscape: the time–height section (24 hours) and the ILS approach. The strips of the ILS view are below the profile; scroll down to see them.

Updates. Only the view on the screen is updated, and nothing while the phone's screen is off or another app is in front; when you come back the view is updated at once. Table 1 gives the intervals. The ILS view is the only one that updates every few seconds: it loads only the aircraft on the approach and the active alerts, compressed — about 1–5 kB each time, depending on the traffic — so watching it for an hour on mobile data uses roughly 1–4 megabytes. The other views use little.

View	Updated	Data
Skew-T · Area now	every 2 min	the area profile; METAR every 10 min
Skew-T · Area hour	when you choose an hour	stored hourly profile
Time–height	every 10 min	hourly profiles of 12 h – 14 days
ILS approach	every 5 s	live approach traffic, alerts; METAR and QNH every 10 min
ACI · WS · ST	every minute	landings, departures, go-arounds of 3 h – 1 day; space–time diagram
History	every minute (not for a date)	landings or departures of 1 h – 1 day or of one date
Health dot	every minute	the system health indicators

Table 1. When the views are updated (only while shown and the screen is on).

4 Skew-T: Area now and Area hour

The Skew-T is the same diagram as the Live Map's *Atmosphere Profile* and the Sounding page (Fig. 4): temperature against pressure, with the isotherms tilted to the right, and the wind as barbs at the right. The grey curve and the grey barbs with their values are the *aircraft* — the average of all aircraft within 150 km. With Sonde the Jokioinen radiosonde is drawn in pink (barbs inside the right edge of the plot), with MEPS the MEPS weather model in teal (barbs in the margin, just left of the aircraft's). The labels at the top left give the data quality, the sounding's and the model's time and how far the aircraft are from them.



Figure 4. Left: Skew-T · Area now, the last 60 minutes, with Sonde (pink) and MEPS (teal) on; the labels give how far the aircraft are from them: on average 0.0 K from the 12 UTC sounding and −0.3 K from the model, wind 10 and 7 kt RMS. Right: Area · hour 24 October 2026 15–16 UTC with Analysis: Levels (tropopause, 0 °C level, maximum wind); a tap at 447 hPa shows the aircraft (−29.7 °C, 191°/35 kt), the sonde and the model at that level.

Area now is the live profile, updated every 2 minutes, with the latest METAR at the ground (ground line, temperature and dewpoint). **Area hour** shows one stored hour: choose the date and the hour in the menu, or step with ⟨ ⟩. The METAR of that hour is stored with it.

Option	Choices	What it does
Window	30 min, 60 min, 2 h, 3 h	period of Area now; longer = more aircraft, less up to date
Date, Hour	calendar, ⟨ ⟩	the stored hour of Area hour
Analysis	Off, Levels, Stability, Turbulence, Advection, Full	automatic marks: tropopause, isotherms, jets, inversions, stability, turbulence risk, warm / cold advection, contrails, surface parcel
Analysis of	Mode S, MEPS, Sonde	which profile the analysis is computed from; MEPS and Sonde add likely cloud layers
Airport profile	Off, Overlay, Zoom	the fine low-level profile of the landings and departures at EFHK; Zoom shows the lowest 5 000 ft
Smoothing	Off, 3 layers, 5 layers	vertical smoothing of the aircraft profile
Sonde	switch	the Jokioinen radiosonde
MEPS	switch	the MEPS weather model over the same area
RH	switch	the trial moisture strip with the sonde's and the model's humidity
Data quality	switch	the data badge and the rings at layers with only 1–2 aircraft

Table 2. Skew-T options. The *Reading the Atmosphere Profile (Skew-T)* guide explains each of them.

5 Time–height

The time–height section puts the hourly profiles side by side: time runs to the right, height up (Fig. 5). The colour is the chosen field, the barbs the hourly wind; lines mark the 0, −10 and −20 °C levels, the tropopause and the maximum wind when switched on. The strip at the bottom gives the aircraft per hour; a pink triangle at the top marks a Jokioinen launch, and the last column (dashed frame) is the hour in progress. Tap a cell for its values.

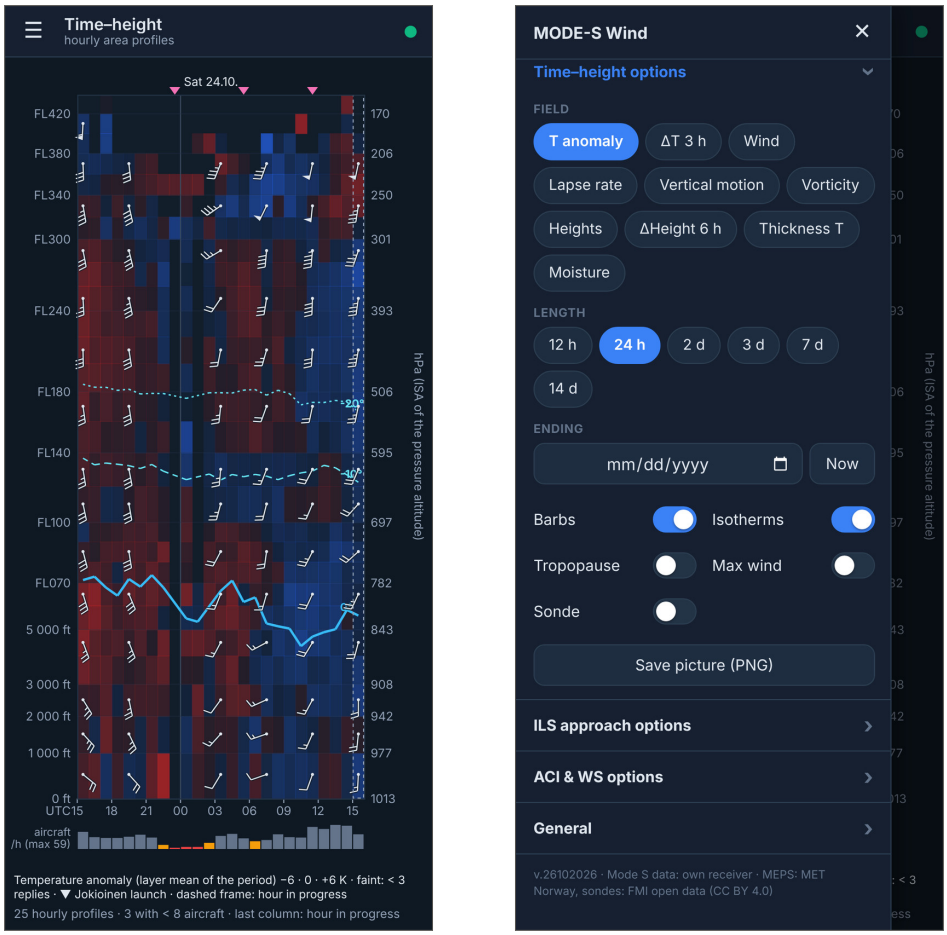


Figure 5. Time–height, 24 hours to 24 October 2026 15 UTC, temperature anomaly (blue colder, red warmer than the period's mean in that layer), with barbs and isotherms; right: its options.

Options. Field: temperature anomaly, temperature change over 3 h, wind speed, lapse rate, vertical motion, vorticity, heights of the pressure surfaces, height change over 6 h, thickness temperature check and the moisture trial. Length: 12 h to 14 days — 24 h by default on a phone; landscape gives the longer periods room. Ending: a past date, or Now. Switches for barbs, isotherms, tropopause, maximum wind and the sonde as its own column. Save picture (PNG) stores the section as an image on the phone. The *Sounding and Wind Map pages* guide explains the fields.

6 ILS approach

The ILS view is the Windshear page's vertical profile of the final approach, live (Fig. 6): distance from the runway threshold to the right (15 NM to the threshold), altitude up (to 5 000 ft). It is redrawn every 5 seconds.

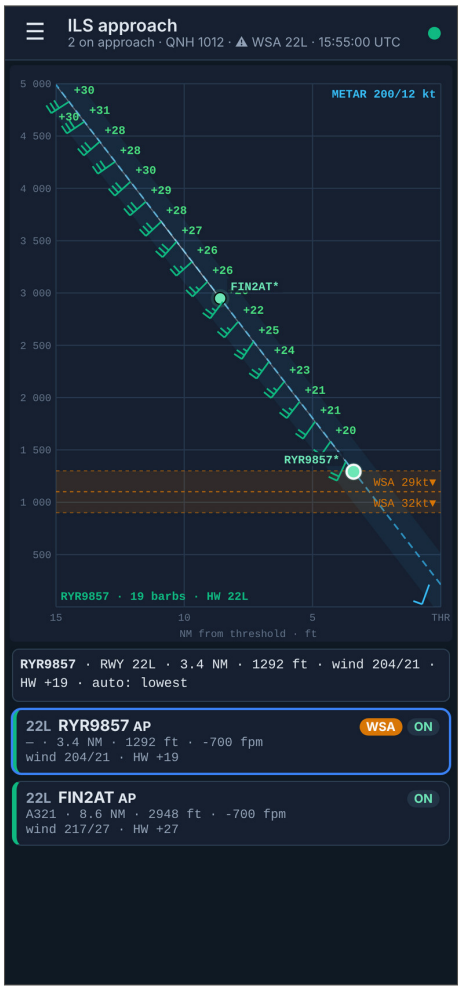


Figure 6. ILS approach, runway 22L. Two aircraft on the glideslope (green `ON`); the barbs are those of the lower one, RYR9857, measured along its approach from 15 NM in, with the headwind component of each in kt above it; a windshear alert (`WSA`) is active at 900–1 300 ft. The METAR surface wind is the blue barb at the threshold. Under the profile the chosen aircraft and the strips. *Illustration with simulated approach traffic.*

What is drawn.

- **The glideslope**: the dashed line is the 3° path (318.5 ft per NM above the threshold, corrected for the QNH), the shaded band ±300 ft around it.
- **Aircraft** in the approach corridors as dots with their callsigns (⋆: autopilot engaged): green *on* the glideslope, amber *high*, red *low*; the faint line behind each is its path over the last minute.
- **Wind barbs** of one aircraft, measured by it on the way down, coloured by where the wind comes from (green: computed from the aircraft's speed and heading replies; blue, orange, violet: other report types). The number above each barb is the headwind component for that runway (Headwind; positive = headwind) or the crosswind (Crosswind). Small hollow circles mark points where no wind could be computed (amber: the aircraft was turning; grey: a gap in its position data). The line at the bottom left names the aircraft and the number of barbs.
- **Windshear alerts**: an active alert is a band across the profile at its altitudes, blue `ADV` (advisory), amber `WSA` (windshear alert) or red `MBA` (microburst alert), with the change in kt (▼ loss, ▲ gain of headwind).
- **METAR**: the surface wind as a blue barb at the threshold, its text at the top right.

Whose barbs. By default the barbs follow the *lowest* aircraft on the approach, the one closest to the threshold — the freshest picture of the wind it is flying through — and change to the next aircraft when it lands. Tap another aircraft on the profile, or its strip, to follow that one; the line under the profile then says `chosen` and an Auto button takes you back to the lowest.

The strips under the profile list the aircraft on the approach, nearest first: runway, callsign (`AP`: autopilot), any windshear alert of that aircraft, the glideslope status; aircraft type, distance, altitude and vertical speed; the latest wind and its headwind (`HW`) or tailwind (`TW`) component. The coloured left edge is the wind source as for the barbs. The top bar gives the count, the QNH used for the glideslope and the active alerts.

Option	Choices	What it does
Runway	All, 04L+R, 22L+R, 04L … 33	the approaches shown
Range	15 NM, 7.5 NM	7.5 NM spreads the last part of the approach
Wind barbs	Lo, Hi, Off	Lo: one barb per 400 ft / 0.5 NM; Hi: per 150 ft / 0.2 NM (denser, more scatter)
Wind component	Headwind, Crosswind, Off	the number above each barb
Windshear alerts	ADV +, WSA +, MBA, Off	the lowest alert level drawn (the same setting as on the Windshear page of the same browser)
Trails	switch	each aircraft's path over the last minute

Table 3. ILS approach options. The *Windshear detection and the Windshear page* guide explains the alerts and the barbs.

7 ACI · WS · ST

The approach conditions and windshear chart of the Windshear page (Fig. 7). The upper part is the *Approach Conditions Index* (ACI, 0–10) of every landing: how rough the last part of the approach was, from the aircraft's own replies — below 2 smooth, 2–4 light, 4–6 choppy, 6–8 rough, 8 and above very rough. The dots are coloured by the aircraft class (`N` narrowbody, `R` regional jet, `T` turboprop, `W` widebody), the line is the running median. With ACI + windshear the lower part shows the windshear report of each landing and departure on the same time axis, as a change of headwind in kt: the strongest headwind ramp of a landing as a stem from zero, the change against the previous 45 minutes as a filled diamond (landings) or a hollow one (departures), go-arounds as triangles, and the advisory and alert levels as dashed lines.

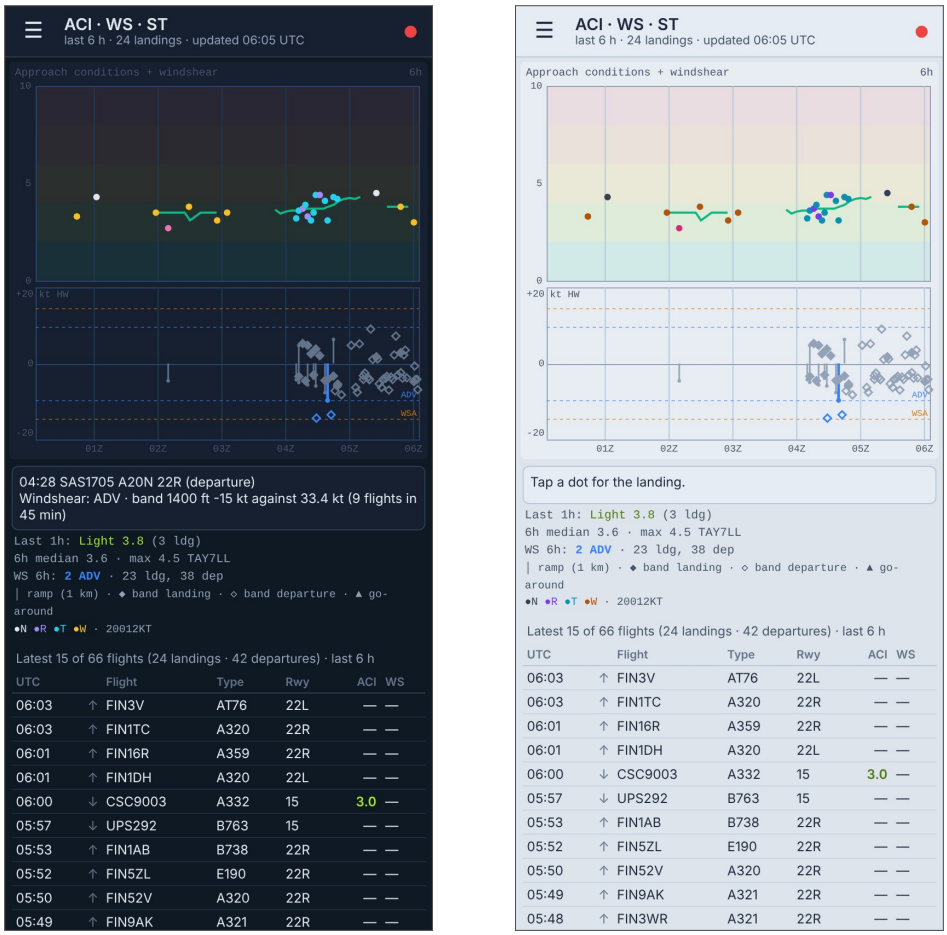


Figure 7. ACI + windshear, the last 6 hours to 28 October 2026 06:05 UTC: landings on runway 15, departures from 22L/22R in a south-westerly low-level jet, ACI mostly 3–4.5. Left: a tap on a diamond shows that flight (here the departure SAS1705, an advisory: 15 kt less headwind at 1 400 ft than the flights before); right: the light theme. The list under the chart is in Fig. 8.

Under the chart: the box with the tapped landing, departure or go-around; the summary lines of the Windshear page (last hour, median and maximum of the period, the windshear alerts and the counts); and the **list of the flights of the period** (from 28 October 2026; Fig. 8), newest first. Its heading gives the count and the period (*Latest 15 of 59 flights*

(17 landings · 42 departures) · last 3 h); 15 rows are shown, Show all opens the rest. The list follows the chart: with ACI only the landings, with ACI + windshear the landings (↓) and departures (↑) — time, flight, aircraft type, runway, ACI (landings only) and windshear level — and with Space–time the flights of the diagram with their largest band change and its height (*−7 @1200*). Landings in History and Departures in History open the History view (Section 8) with the same period, for the bands and reports of each flight. **Options**: the period (3 h, 6 h, 12 h, 1 day) and ACI + windshear, ACI only or Space–time. The *Windshear* guide and the *Approach Conditions Index* article explain them.

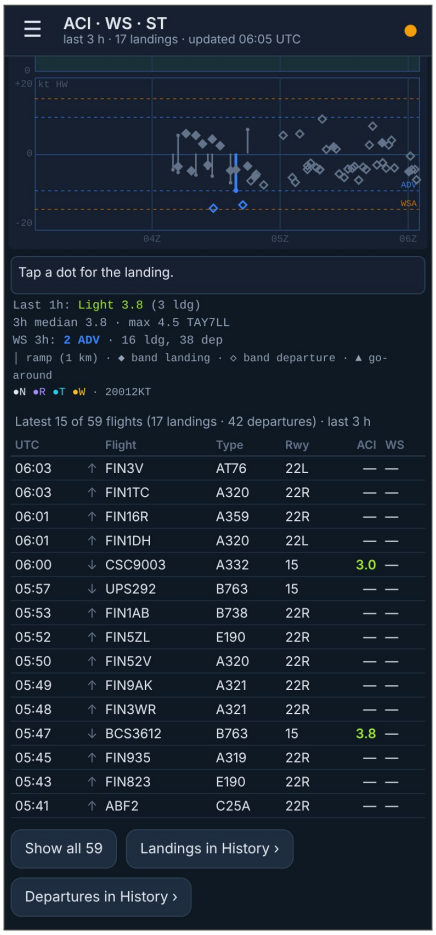


Figure 8. The list under the chart (scrolled down), ACI + windshear, the last 3 hours to 28 October 2026 06:05 UTC: the latest 15 of 59 flights — landings on runway 15 (↓) and departures from 22L/22R (↑) — with Show all 59 and the History buttons.

Space–time (from 28 October 2026; Fig. 9) shows how the wind changes move through the successive flights: time to the right, height (400–1 800 ft) upwards, and in each cell the median change of the headwind of the landings and departures in that 5–30 minutes against the flights of the previous 45 minutes — **blue** less headwind, **red** more, from 3 to 15 kt. A front or a shear layer is a coloured streak through several flights and bands, turbulence scattered single cells. A dot marks a flight at or beyond its advisory limit, a triangle at the top a flight with a windshear level. Tap a cell: the box below shows the cell, and the list its flights with their change and wind (All flights × goes back to the whole period). Turn the phone for shorter time cells.

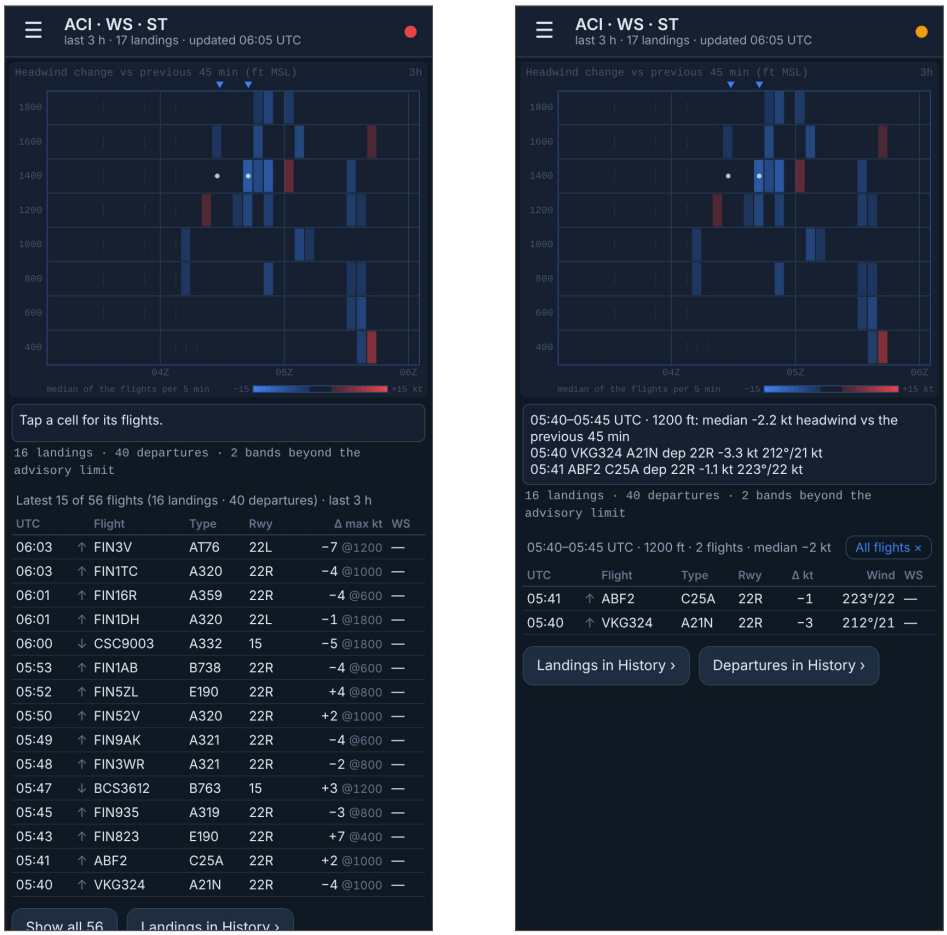


Figure 9. Space–time, the last 3 hours to 28 October 2026 06:05 UTC: the morning departures from 22L/22R climbing into a weakening south-westerly low-level jet — blue cells (less headwind) at 1 200–1 600 ft. Left: the diagram and its flights, each with its largest band change; right: a tapped cell (05:40–05:45 UTC, 1 200 ft) and its two departures in the list.

8 History

The Approach History of the Windshear page made for a phone (Fig. 10): the **landings** or the **departures**, newest first, as cards.

- **Line 1**: time (UTC; earlier days with the date), callsign, aircraft type, runway (in a thin frame; the columns line up from card to card), and chips: the **ACI** value, the **windshear level** (`ADV`, `WSA`, `MBA`), ***trial*** (a departure whose headwind ramps along the climb would have given that level — a trial, no alert) and **GA** (go-around).
- **Line 2**: depends on Show. Headwind and Crosswind: one cell per height band (the heights are in the row under the search field) with the component along the runway in kt — headwind green, tailwind red; crosswind + from the right, − from the left. Wind: an arrow (where the wind blows to) and the speed. ACI: the index and its two segments with the METAR. Windshear: the strongest headwind ramp and the band change.
- **Tap a card** to open it: all its height bands with wind, headwind and crosswind, the approach conditions in detail, and the windshear report in words — the texts the full site shows when the mouse is over a row. Tap again to close.

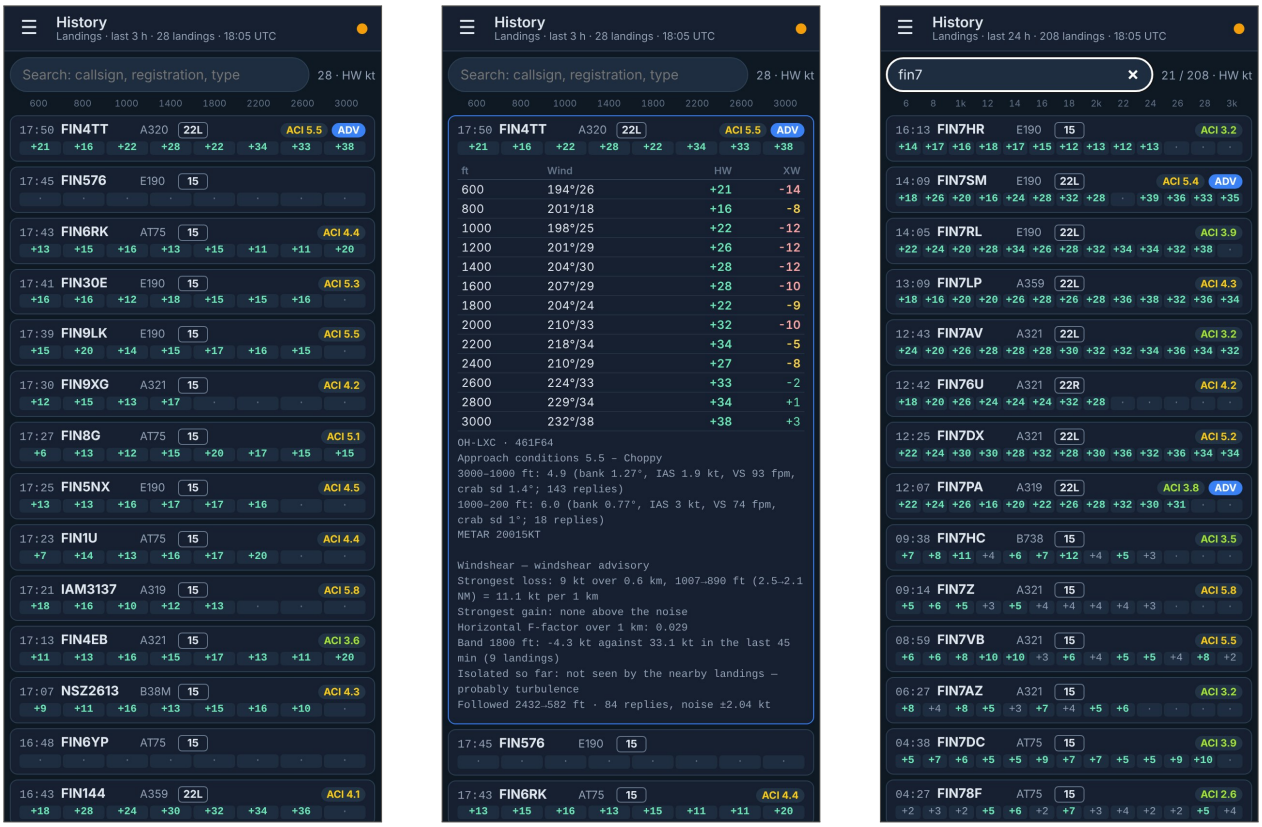


Figure 10. History, 27 October 2026. Left: the landings of the last 3 hours, headwind per band (a strong south-westerly: +13 to +38 kt). Middle: a card opened — the bands, the approach conditions and the windshear report. Right: the search `fin7` over a day, with the 13 height bands of Hi.

Search. The field above the cards finds the callsign, registration or aircraft type anywhere in the text, upper or lower case and with or without the hyphen: `fin7`, `OH-LX` or `ohlx`, `AT75`. The number beside it shows how many of the flights match. The search on the full site's Approach History works the same way.

Options (Fig. 11): Landings / Departures; the range 1 h – 1 day (default 3 h); or a **date**: the calendar field or ⟨ ⟩ (a day back / forward), and Live returns to the range; Show; and Lo · 8 or Hi · 13 height bands. A live range is updated every minute; a date is loaded once.

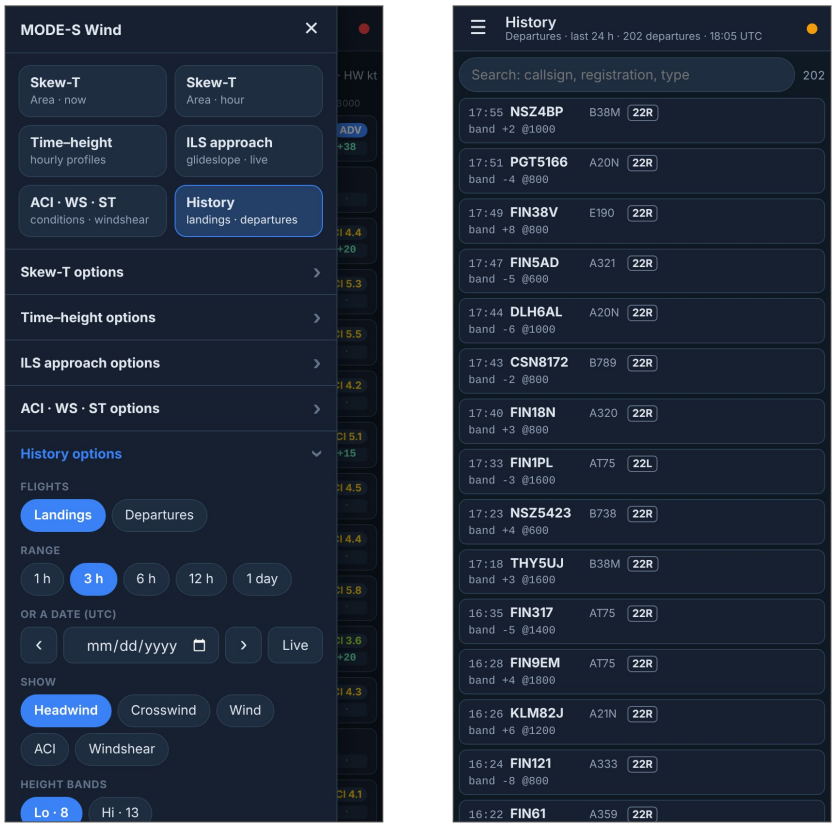


Figure 11. Left: the History options. Right: the departures of a day, Windshear shown — the band change of each climb-out against the earlier flights.

9 When something looks wrong

- **The page does not load, or asks for the password again and again.** The phone must reach MODE-S Wind as a computer does; check the network and the address (with `/app`). A home-screen icon can lose the saved login after an update of the phone; type it again.
- **`Could not load ...`** under a view: the server did not answer in time. The view tries again at its next update, or switch the view and back.
- **The health dot is amber or red.** Open the menu: General says which part. MODE-S Wind keeps working with what it has — e.g. without new MEPS data the Skew-T has no model line, the rest is unchanged.
- **`No aircraft on the approach`**: there is no traffic in the approach corridors of the chosen runway; choose All.
- **`MEPS: no model data for this time`** or `no sounding stored`: the model run or the radiosonde for that time has not arrived (or is not available); the aircraft profile is not affected.
- **After an update of MODE-S Wind** the phone may still show the old page from its cache: reload the page in the browser, or close the home-screen app and open it again.
- **Text too small?** Use landscape, or the phone's own zoom (pinch); the diagrams are drawn at the phone's full resolution.

10 Quick reference

Item	Meaning
`/app`	the mobile view; same login as the full site
≡	menu: views as tiles, options of each product, General
Dot at the top right	system health: green working, amber late / fallback, red stopped
Tap / drag	values at that point (the mouse hover of the full site)
Swipe up / down	scroll a view taller than the screen
Grey line and barbs	aircraft profile (Skew-T), with the wind values
Pink / teal	Jokioinen radiosonde / MEPS weather model
Green / amber / red dot (ILS)	aircraft on / above / below the glideslope (±300 ft)
`ADV` / `WSA` / `MBA`	windshear advisory / windshear alert / microburst alert
`HW` / `TW` / `AP`	headwind / tailwind component; autopilot engaged
ACI	Approach Conditions Index 0–10: below 2 smooth, light, choppy, rough, 8+ very rough
Updates	Skew-T now 2 min, time–height 10 min, ILS 5 s, ACI · WS · ST and History 1 min; only the shown view, only with the screen on
Blue / red cells (ST)	less / more headwind than the flights of the previous 45 min
Card chips (History)	ACI value, windshear level, *trial* (departure ramps, no alert), GA go-around
Settings	remembered on the phone; Skew-T options and theme shared with the full site in the same browser

Other guides (the User guides tab of the full site's *What's new* window): *Reading the Atmosphere Profile (Skew-T)*; *The Sounding and Wind Map pages*; *Windshear detection and the Windshear page*; *Mode S & ADS-B and the Statistics page*.